# BAB 2. TCP/IP Model

## 2.1 Tujuan

- Mahasiswa mampu melakukan identifikasi transmisi data menggunakan model TCP/IP
- Mahasiswa mampu melakukan identifikasi layer dari model TCP/IP
- Mahasiswa mampu menggunakan aplikasi untuk mengidentifikasi data dari model TCP/IP

## 2.2 Dasar Teori

### 2.2.1 OSI Layer

OSI adalah referensi komunikasi dari Open System Interconnection. OSI model digunakan sebagai referensi untuk membahas spesifikasi protokol. OSI model terdiri dari 7 layer seperti pada tabel 1.

| Layer # | Nama | Unit Enkapsulasi | Deskripsi/Fungsi |
|---|---|---|---|
| 7 | Application | Data | Mendefinisikan pelayanan (services) komunikasi pada suatu aplikasi |
| 6 | Presentation | Data | Mendefinisikan format data |
| 5 | Session | Data | Mendefinisikan memulai, pengontrolan dan mengakhiri komunikasi |
| 4 | Transport | Segment | Mendefinisikan jenis pengiriman |
| 3 | Network | Paket | Mendefinisikan pengiriman dari titik ke titik. |
| 2 | Data Link | Frame | Mendefinisikan cara pengaturan pengiriman data |
| 1 | Physical | bit | Mendifinisikan bentuk interface dari sebuah media transmisi |

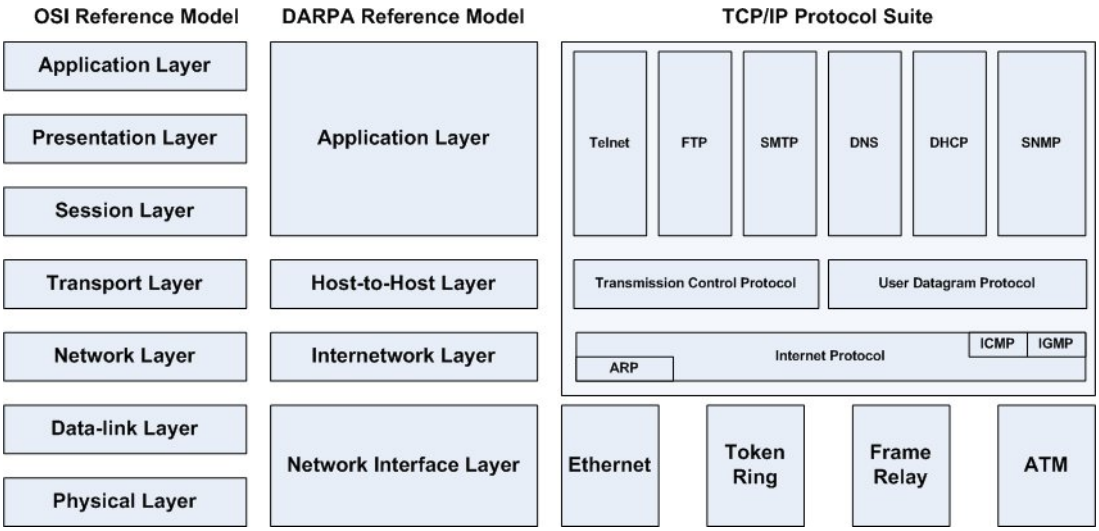
**Tabel 1. OSI Layer**

Dimana setiap layer memiliki fungsi dan contoh masing-masing.

### 2.2.2 TCP/IP Model

TCP/IP (singkatan dari Transmission Control Protocol/Internet Protocol), adalah gabungan dari protokol TCP dan IP sebagai sekelompok protokol yang mengatur komunikasi data dalam proses tukar-menukar data dari satu komputer ke komputer lain di dalam jaringan Internet yang akan memastikan pengiriman data sampai ke alamat yang dituju. Protokol ini tidaklah dapat berdiri sendiri, karena memang protokol ini berupa kumpulan protokol (Protocol Suite).

Arsitektur TCP/IP tidaklah berbasis OSI Layer, tetapi menggunakan model referensi DARPA. Seperti pada gambar 11., TCP/IP mengimplementasikan arsitektur berlapis yang terdiri atas empat lapis. Empat lapis ini, dapat dipetakan terhadap model OSI layer. Empat lapis ini, disebut juga DARPA Model atau DOD Model.




**Gambar 11. OSI Layer vs TCP/IP Protocol Suite.**

## 2.3 Pengenalan Aplikasi

### 2.3.1 Wireshark

Wireshark adalah aplikasi penganalisa paket jaringan (Network Packet Analyzer). Sebuah software penganalisa paket jaringan yang bekerja dengan cara menangkap paket jaringan dan mencoba untuk menampilkan data paket yang tertangkap dengan detail.

Wireshark bersifat open source.

Contoh penggunaan wireshark:
- Administrator jaringan menggunakan untuk melakukan penangan masalah jaringan
- Network security engineer menggunakan untuk memeriksa keamanan jaringan
- Developer menggunakan untuk melakukan debug pada implementasi protokol
- Umum digunakan untuk mempelajari internal protokol jaringan

Wireshark memiliki fitur:
- Tersedia untuk UNIX dan Windows.
- Menangkap paket data langsung dari network interface.
- Tampilan paket dengan informasi protokol yang sangat rinci.
- Open dan Save data paket yang dicapture.
- Import dan Eksport paket data dari dank e banyak program capture lainnya.
- Menyaring paket data dengan berbagai macam kriteria.
- Warnai layar paket berdasarkan filter.

### 2.3.2 Cara penggunaan Wireshark

Langkah-langkah menggunakan wireshark

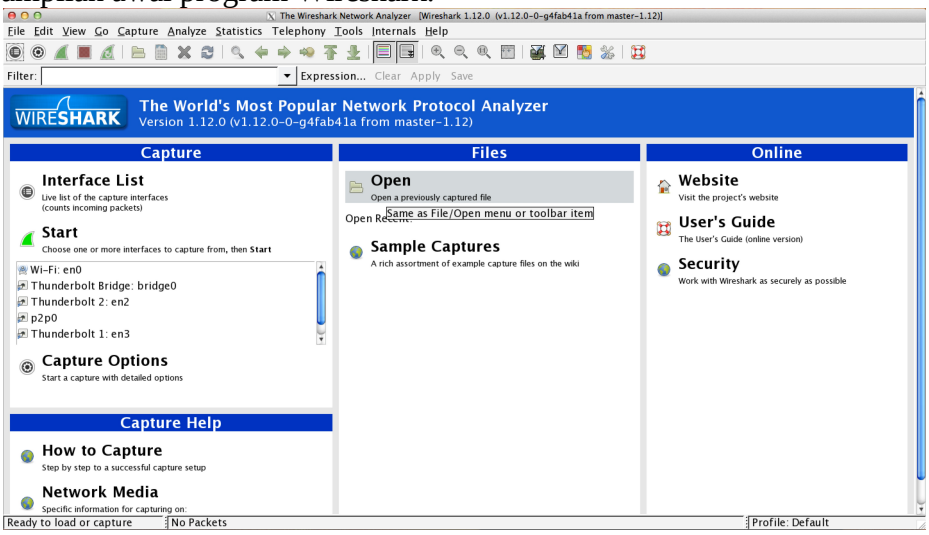
1. Klik ikon…



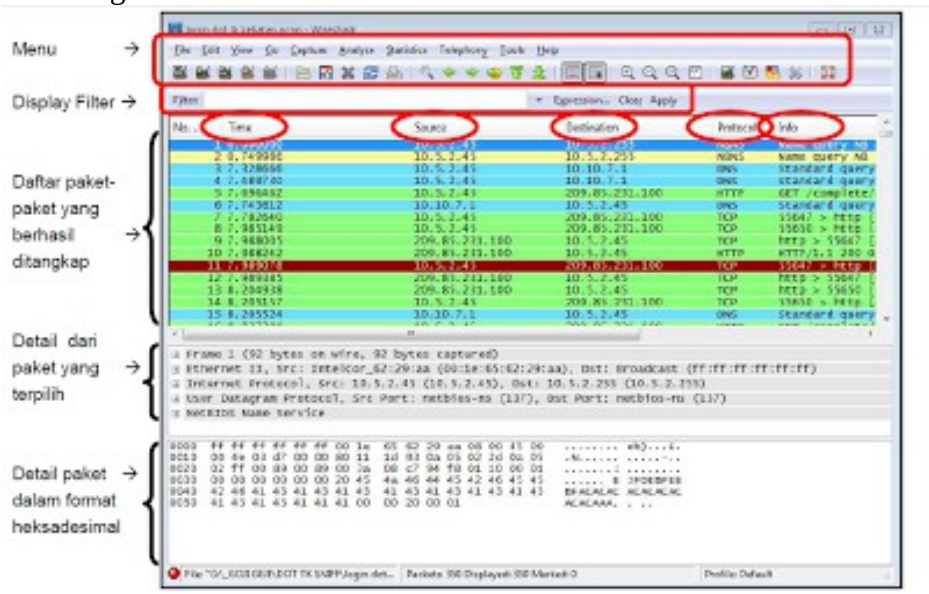
2. Kemudian akan tampil aplikasi wireshark dengan diawali spash screen.



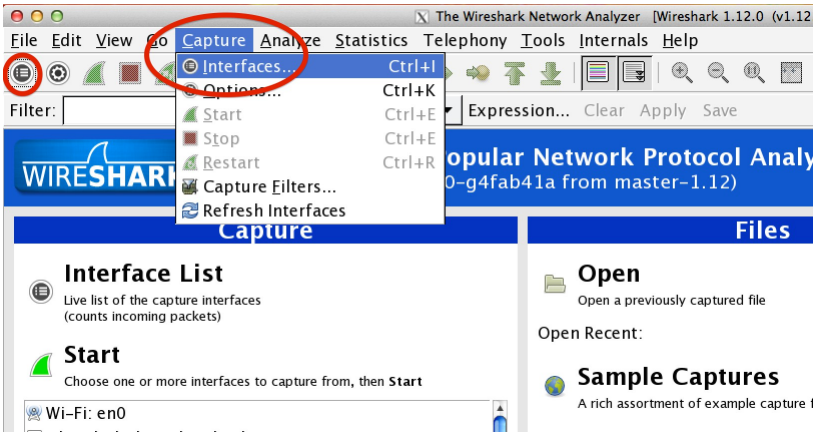

3. Tampilan awal program Wireshark.



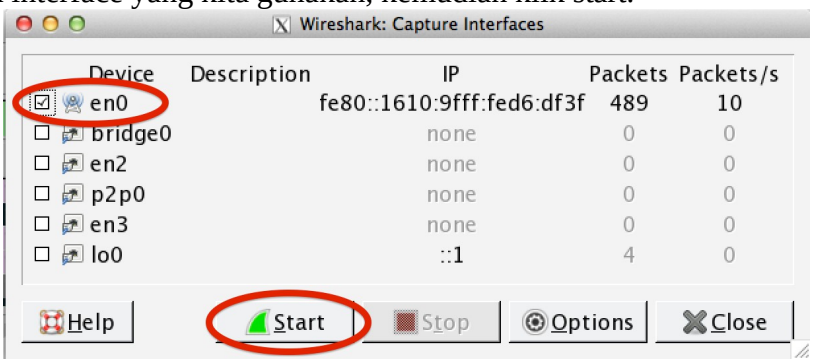

4. Keterangan menu wireshark.



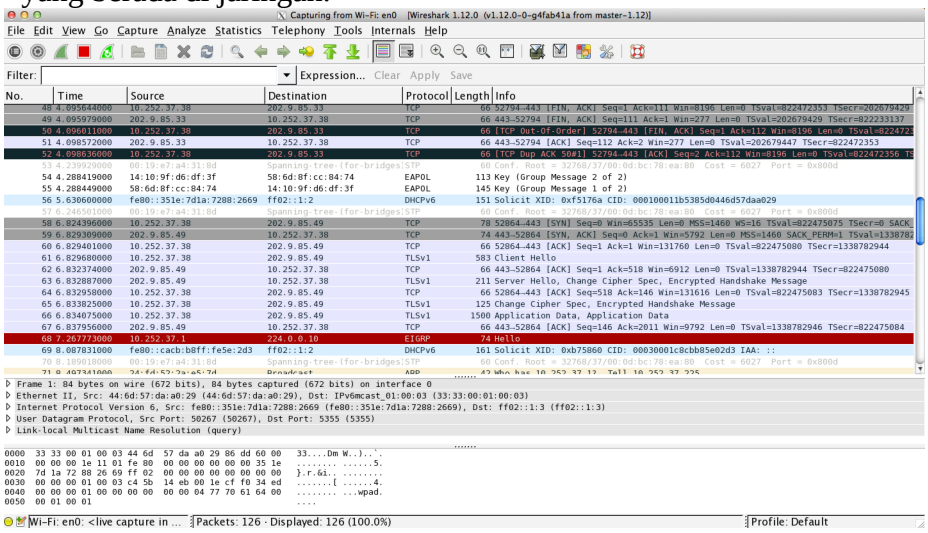

5. Melakukan CAPTURE data menggunakan wireshark. Pilih menu INTERFACE.




6. Pilih interface yang kita gunakan, kemudian klik start.



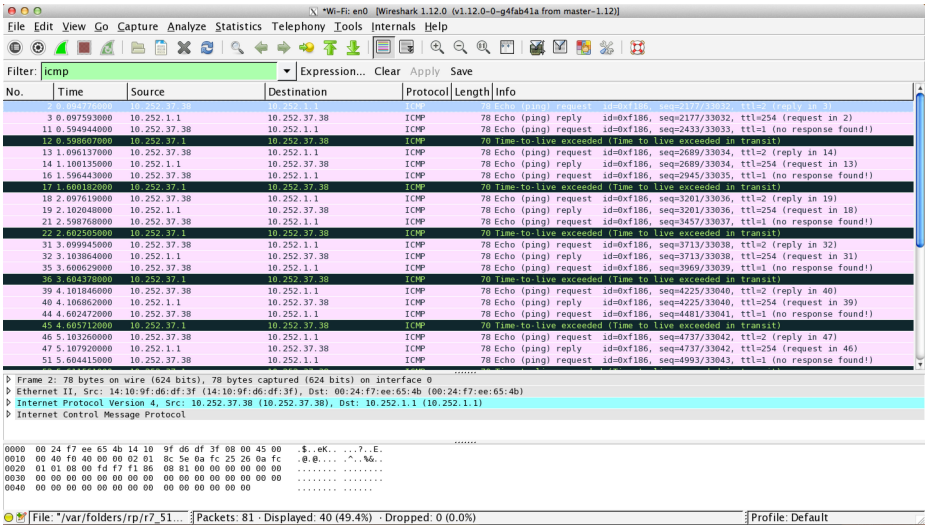

7. Setelah mengaktifkan interfaces, Wireshark akan beraksi membaca semua data yang berada di jaringan.



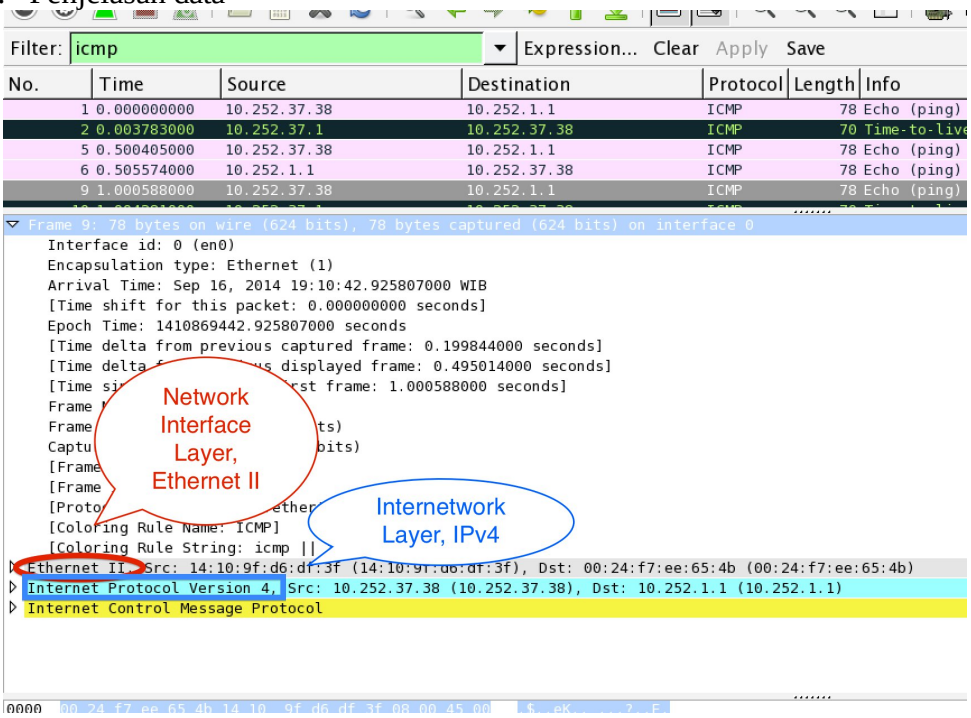

Terlihat banyak data yang dapat dicapture.
Keterangan:

| | |
|---|---|
| Source | Alamat IP pengirim |
| Destination | Alamat IP tujuan |
| Protocol | Jenis protokol yang digunakan, seperti TCP, DHCP, dll |
| Length | Panjang data yang dikirimkan |
| Info | Informasi isi data yang dikirimkan |

8. Ketikkan ICMP pada filter box untuk melakukan penyaringan data yang akan ditampilkan sehingga hanya data PING yang akan ditampilkan.



9. Penjelasan data




## 2.4 Persiapan Praktikum

1. Modul praktikum
2. PC yang dilengkapi dengan aplikasi Wireshark
3. Jaringan aktif
4. Lembar Laporan Praktikum (Lampiran)
5. Koneksi LAN dan Internet

## 2.5 Langkah-langkah Praktikum

### 2.5.1 Menjalankan aplikasi wireshark

1. Mahasiswa melakukan pengecekan apakah pada PC yang dipakai sudah terinstall aplikasi wireshark ?!!

2. Apabila belum terinstall, mahasiswa harus melakukan installasi di PC yang digunakan !!!

   Untuk OS Linux Debian/Ubuntu:
   dhoto@mypc $ su
   root@mypc # apt-get install wireshark

   Untuk OSX:
   - Download wireshark dan XQuartz
   - Lakukan konfigurasi XQuartz pada wireshark

   Untuk Windows:
   - Download wireshark
   - Klik 2x pada installer wireshark

3. Klik ikon Wireshark !!!
4. Klik pada menu CAPTURE dan memilih INTERFACES.
5. Pilih interface yang akan digunakan untuk mengrekam data.
6. Klik Start.
7. Pada OS linux, jalankan aplikasi TERMINAL dan ketik perintah PING yang digunakan untuk melakukan pengecekan jaringan (tanpa tanda $):
   $ ping 10.252.1.1
8. Kembali ke aplikasi wireshark, hentikan rekam data dengan cara klik tombol merah pada wireshark. Ambil data untuk jenis tipe: icmp dengan cara memfilter di aplikasi wireshark dan lakukan screen shot pada tampilan wireshark yang kalian gunakan.

### 2.5.2 Melakukan rekam data untuk jenis protokol tertentu

1. Lakukan praktikum seperti kegiatan 2.5.1 dari nomer 4 s/d 8.
2. Ganti kegiatan pada no 7, dengan kegiatan berikut:
   a. Jalankan aplikasi terminal dan ketik perintah NSLOOKUP yang digunakan untuk mengetahui IP dari suatu tujuan (tanpa tanda $):
      $ nslookup www.pens.ac.id
      Ambil data untuk jenis tipe: dns
   b. Jalankan aplikasi web browser dan arahkan ke alamat:
      ftp://newfs.pens.ac.id
      Download salah satu data yang ada di alamat tersebut !!!
      Ambil data untuk jenis tipe: ftp
   c. Jalankan aplikasi web browser dan arahkan ke alamat:
      http://www.pens.ac.id
      Ambil data untuk jenis tipe: http
3. Untuk masing-masing kegiatan diatas, laporkan data yang telah direkam sesuai dengan layer pada TCP/IP model. Gunakan lembar **Laporan praktikum 2. TCP/IP Model**. (Lampiran)

### 2.6.1 Lampiran

**Laporan praktikum 2. TCP/IP Model**

Kegiatan : ping IP 10.252.1.1 (ICMP)
Nama : Mazda Maulana
NRP : 5325600057

**Data layer**

Application layer

```
Internet Control Message Protocol
    Type: 0 (Echo (ping) reply)
    Code: 0
    Checksum: 0xe575 [correct]
    [Checksum Status: Good]
    Identifier (BE): 62215 (0xf307)
    Identifier (LE): 2035 (0x07f3)
    Sequence Number (BE): 1 (0x0001)
    Sequence Number (LE): 256 (0x0100)
    [Request frame: 2516]
    [Response time: 5.731 ms]
    Timestamp from icmp data: Oct 20, 2026 14:50:30.271943000 WIB
    [Timestamp from icmp data (relative): 0.005803921 seconds]
    Data (40 bytes)
        Data:
101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f303132333435363
7
        [Length: 40]
```

Host-to-host layer

Internetwork layer

```
Internet Protocol Version 4, Src: 10.252.1.1, Dst: 10.252.133.128
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 84
    Identification: 0xeeb5 (61109)
    010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
        .1.. .... = Don't fragment: Set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 253
    Protocol: ICMP (1)
    Header Checksum: 0xf279 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 10.252.1.1
    Destination Address: 10.252.133.128
```

Network Interface layer

```
Ethernet II, Src: VMware_9f:e1:85 (00:50:56:9f:e1:85), Dst: HonHaiPrecis_f3:fd:ff
(d8:5d:e2:f3:fd:ff)
    Destination: HonHaiPrecis_f3:fd:ff (d8:5d:e2:f3:fd:ff)
        Address: HonHaiPrecis_f3:fd:ff (d8:5d:e2:f3:fd:ff)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: VMware_9f:e1:85 (00:50:56:9f:e1:85)
```

```
    Address: VMware_9f:e1:85 (00:50:56:9f:e1:85)
    .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
    .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
```

**2.6.2**

**Laporan praktikum 2. TCP/IP Model**

Kegiatan : NSLOOKUP www.pens.ac.id (DNS)
Nama : Mazda Maulana
NRP : 5325600057

**Data layer**

Application layer

```
Domain Name System (query)
  Transaction ID: 0xfc5c
  Flags: 0x0100 Standard query
    0... .... .... .... = Response: Message is a query
    .000 0... .... .... = Opcode: Standard query (0)
    .... ..0. .... .... = Truncated: Message is not truncated
    .... ...1 .... .... = Recursion desired: Do query recursively
    .... .... .0.. .... = Z: reserved (0)
    .... .... ...0 .... = Non-authenticated data: Unacceptable
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  Queries
    www.pens.ac.id: type A, class IN
      Name: www.pens.ac.id
      [Name Length: 14]
      [Label Count: 4]
      Type: A (1) (Host Address)
      Class: IN (0x0001)
  [Response In: 2070]
```

Host-to-host layer

```
User Datagram Protocol, Src Port: 47957, Dst Port: 53
  Source Port: 47957
  Destination Port: 53
  Length: 40
  Checksum: 0x109c [unverified]
  [Checksum Status: Unverified]
  [Stream index: 307]
  [Timestamps]
    [Time since first frame: 0.000000000 seconds]
    [Time since previous frame: 0.000000000 seconds]
  UDP payload (32 bytes)
```

Internetwork layer

```
Internet Protocol Version 4, Src: 10.252.128.129, Dst: 10.252.128.2
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 60
  Identification: 0xd30a (54026)
  000. .... = Flags: 0x0
```

```
    0... .... = Reserved bit: Not set
    .0.. .... = Don't fragment: Not set
    ..0. .... = More fragments: Not set
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 64
  Protocol: UDP (17)
  Header Checksum: 0x912b [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 10.252.128.129
  Destination Address: 10.252.128.2
```

Network Interface layer

```
Ethernet II, Src: HonHaiPrecis_f3:fd:ff (d8:5d:e2:f3:fd:ff), Dst: VMware_9f:5a:c2
(00:50:56:9f:5a:c2)
  Destination: VMware_9f:5a:c2 (00:50:56:9f:5a:c2)
    Address: VMware_9f:5a:c2 (00:50:56:9f:5a:c2)
    .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
    .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Source: HonHaiPrecis_f3:fd:ff (d8:5d:e2:f3:fd:ff)
    Address: HonHaiPrecis_f3:fd:ff (d8:5d:e2:f3:fd:ff)
    .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
    .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
```

### 2.6.3 Lampiran

**Laporan praktikum 2. TCP/IP Model**

Kegiatan : http://www.pens.ac.id (HTTP)
Nama : Mazda Maulana
NRP : 5325600057

**Data layer**

Application layer

```
Hypertext Transfer Protocol
  HTTP/1.1 301 Moved Permanently\r\n
    [Expert Info (Chat/Sequence): HTTP/1.1 301 Moved Permanently\r\n]
      [HTTP/1.1 301 Moved Permanently\r\n]
      [Severity level: Chat]
      [Group: Sequence]
    Response Version: HTTP/1.1
    Status Code: 301
    [Status Code Description: Moved Permanently]
    Response Phrase: Moved Permanently
  Server: nginx\r\n
  Date: Tue, 13 Oct 2026 09:20:51 GMT\r\n
  Content-Type: text/html\r\n
  Content-Length: 162\r\n
    [Content length: 162]
  Connection: keep-alive\r\n
  Location: https://www.pens.ac.id/\r\n
  \r\n
```

```
    [HTTP response 1/1]
    [Time since request: 0.068317732 seconds]
    [Request in frame: 66624]
    [Request URI: http://www.pens.ac.id/]
    File Data: 162 bytes
Line-based text data: text/html (7 lines)
    <html>\r\n
    <head><title>301 Moved Permanently</title></head>\r\n
    <body>\r\n
    <center><h1>301 Moved Permanently</h1></center>\r\n
    <hr><center>nginx</center>\r\n
    </body>\r\n
    </html>\r\n
```

Host-to-host layer

```
Transmission Control Protocol, Src Port: 80, Dst Port: 48884, Seq: 1, Ack: 343, Len: 353
    Source Port: 80
    Destination Port: 48884
    [Stream index: 67]
    [Conversation completeness: Complete, WITH_DATA (31)]
        ..0. .... = RST: Absent
        ...1 .... = FIN: Present
        .... 1... = Data: Present
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
        [Completeness Flags: ·FDASS]
    [TCP Segment Len: 353]
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 3573448455
    [Next Sequence Number: 354    (relative sequence number)]
    Acknowledgment Number: 343    (relative ack number)
    Acknowledgment number (raw): 4150295195
    1000 .... = Header Length: 32 bytes (8)
    Flags: 0x018 (PSH, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 1... = Push: Set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····AP···]
    Window: 507
    [Calculated window size: 64896]
    [Window size scaling factor: 128]
    Checksum: 0x28fa [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
        TCP Option - No-Operation (NOP)
            Kind: No-Operation (1)
        TCP Option - No-Operation (NOP)
            Kind: No-Operation (1)
        TCP Option - Timestamps
            Kind: Time Stamp Option (8)
            Length: 10
            Timestamp value: 1651449509: TSval 1651449509, TSecr 1381339798
            Timestamp echo reply: 1381339798
    [Timestamps]
```

```
    [Time since first frame in this TCP stream: 0.107488613 seconds]
    [Time since previous frame in this TCP stream: 0.000027518 seconds]
  [SEQ/ACK analysis]
    [iRTT: 0.038340133 seconds]
    [Bytes in flight: 353]
    [Bytes sent since last PSH flag: 353]
  TCP payload (353 bytes)
```

Internetwork layer

```
Internet Protocol Version 4, Src: 202.9.85.176, Dst: 10.252.128.129
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 405
  Identification: 0x7e21 (32289)
  010. .... = Flags: 0x2, Don't fragment
    0... .... = Reserved bit: Not set
    .1.. .... = Don't fragment: Set
    ..0. .... = More fragments: Not set
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 63
  Protocol: TCP (6)
  Header Checksum: 0x110b [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 202.9.85.176
  Destination Address: 10.252.128.129
```

Network Interface layer

```
Ethernet II, Src: VMware_b5:55:03 (00:50:56:b5:55:03), Dst: HonHaiPrecis_f3:fd:ff
(d8:5d:e2:f3:fd:ff)
  Destination: HonHaiPrecis_f3:fd:ff (d8:5d:e2:f3:fd:ff)
    Address: HonHaiPrecis_f3:fd:ff (d8:5d:e2:f3:fd:ff)
    .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
    .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Source: VMware_b5:55:03 (00:50:56:b5:55:03)
    Address: VMware_b5:55:03 (00:50:56:b5:55:03)
    .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
    .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
```

### 2.6.4 Lampiran

**Laporan praktikum 2. TCP/IP Model**

Kegiatan : ftp://newfs.pens.ac.id (FTP)
Nama : Mazda Maulana
NRP : 532600057

**Data layer**

Application layer

```
File Transfer Protocol (FTP)
  150 Opening BINARY mode data connection for 1.4 Hacking.mp4 (41326113 bytes)\r\n
    Response code: File status okay; about to open data connection (150)
    Response arg: Opening BINARY mode data connection for 1.4 Hacking.mp4 (41326113
bytes)
[Current working directory: /ebook/Certified Ethical Hacker (CEH) v12/Module 1 - Introduction
To Ethical Hacking]
```

Host-to-host layer

```
Transmission Control Protocol, Src Port: 21, Dst Port: 61700, Seq: 604, Ack: 289, Len: 78
    Source Port: 21
    Destination Port: 61700
    [Stream index: 20]
    [Stream Packet Number: 45]
    [Conversation completeness: Incomplete, DATA (15)]
        ..0. .... = RST: Absent
        ...0 .... = FIN: Absent
        .... 1... = Data: Present
        .... .1.. = ACK: Present
        .... ..1. = SYN-ACK: Present
        .... ...1 = SYN: Present
        [Completeness Flags: ··DASS]
    [TCP Segment Len: 78]
    Sequence Number: 604    (relative sequence number)
    Sequence Number (raw): 721770193
    [Next Sequence Number: 682    (relative sequence number)]
    Acknowledgment Number: 289    (relative ack number)
    Acknowledgment number (raw): 604470562
    0101 .... = Header Length: 20 bytes (5)
    Flags: 0x018 (PSH, ACK)
        000. .... .... = Reserved: Not set
        ...0 .... .... = Accurate ECN: Not set
        .... 0... .... = Congestion Window Reduced: Not set
        .... .0.. .... = ECN-Echo: Not set
        .... ..0. .... = Urgent: Not set
        .... ...1 .... = Acknowledgment: Set
        .... .... 1... = Push: Set
        .... .... .0.. = Reset: Not set
        .... .... ..0. = Syn: Not set
        .... .... ...0 = Fin: Not set
        [TCP Flags: ·····AP···]
    Window: 229
    [Calculated window size: 29312]
    [Window size scaling factor: 128]
    Checksum: 0x794e [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
    [Timestamps]
        [Time since first frame in this TCP stream: 1 minute, 34.968832600 seconds]
        [Time since previous frame in this TCP stream: 11.400 microseconds]
    [SEQ/ACK analysis]
        [iRTT: 103.337900 milliseconds]
        [Bytes in flight: 78]
        [Bytes sent since last PSH flag: 78]
    [Client Contiguous Streams: 1]
    [Server Contiguous Streams: 1]
    TCP payload (78 bytes)
```

Internetwork layer

```
Internet Protocol Version 4, Src: 202.9.85.80, Dst: 10.252.131.107
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 118
    Identification: 0x7956 (31062)
    010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
```

```
    .1.. .... = Don't fragment: Set
    ..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 63
Protocol: TCP (6)
Header Checksum: 0x146b [validation disabled]
[Header checksum status: Unverified]
Source Address: 202.9.85.80
Destination Address: 10.252.131.107
[Stream index: 25]
```

Network Interface layer

```
Ethernet II, Src: VMware_b5:55:03 (00:50:56:b5:55:03), Dst: LiteonTechno_4b:d5:dd
(24:b2:b9:4b:d5:dd)
  Destination: LiteonTechno_4b:d5:dd (24:b2:b9:4b:d5:dd)
    .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
    .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Source: VMware_b5:55:03 (00:50:56:b5:55:03)
    .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
    .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 40]
```